# A FINITE PRESENTATION MORPHISM OF ALGEBRAIC SPACES OVER A NOETHERIAN BASE HAS NOETHERIAN SOURCE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is a verification of the three conditions of the definition; each step names the lemma or definition it uses, and no step is left to the reader.

*Editorial note.* The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

*Allowed standard prerequisites (editorial).*

- A locally finite type morphism to a locally Noetherian base has locally Noetherian source. Lemmas lemma-finite-presentation-finite-type and lemma-locally-finite-type-locally-noetherian, used at proof lines 5771-5774. Gives the first of the three defining conditions.
- Finite presentation makes a morphism quasi-compact and quasi-separated. Definition definition-locally-finite-presentation, used at proof lines 5776-5777. Combined with quasi-compactness of Y gives quasi-compactness of X.
- Noetherianness of an algebraic space in terms of the three conditions. Properties of Spaces, Definition spaces-properties-definition-noetherian, used at proof lines 5778-5786. Is the definition whose three conditions the proof verifies.
- Separatedness over a quasi-separated base is stable. Lemma lemma-separated-over-separated, used at proof lines 5781-5782. Gives quasi-separatedness of X.

*Source reference.* Original definition 03XP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5156–5156. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 0464: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5459–5459. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 04ZK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6468–6468. Navigation pointer into the Stacks

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*Source reference.* Original lemma 04ZH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6465–6465. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original definition 03EA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4449–4449. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. If $f$ is of finite presentation and $Y$ is Noetherian, then $X$ is Noetherian.*

*Proof.* Assume $f$ is of finite presentation and $Y$ Noetherian. By Lemmas 0464 and 04ZK we see that $X$ is locally Noetherian. As $f$ is quasi-compact and $Y$ is quasi-compact we see that $X$ is quasi-compact. As $f$ is of finite presentation it is quasi-separated (see Definition 03XP) and as $Y$ is Noetherian it is quasi-separated (see Properties of Spaces, Definition 03EA). Hence $X$ is quasi-separated by Lemma 04ZH. Hence we have checked all three conditions of Properties of Spaces, Definition 03EA and we win. □